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Understanding the impact of cycloalkane additives on the combustion of HEFA jet fuel

Alka Panda^{a,*}, Andrew Klingberg^{a,1}, Ronald K. Hanson

^aDepartment of Mechanical Engineering, Stanford University, Stanford, CA 94305, USA

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ABSTRACT

Drop-in biofuels, such as Hydroprocessed Esters and Fatty Acids (HEFA), are designed to deliver performance comparable to petroleum-based jet fuels without requiring modifications to existing aircraft engines. These biofuels, which are primarily n- and isoalkanes, have been certified by ASTM for use in blends of up to 50% with conventional Jet A to take advantage of the physical properties of cycloalkanes and aromatics. Cycloalkanes and aromatics are integral components of conventional jet fuels, contributing to desirable physical and combustion properties. However, aromatics are both carcinogenic and major precursors to soot formation, prompting the need for safer and more sustainable alternatives. Bio-derived cycloalkanes have emerged as promising aromatic substitutes, offering comparable fuel properties while mitigating environmental and health risks. HEFA fuels provide an ideal platform for investigating how variations in cycloalkane structures (e.g., monosubstituted, polysubstituted, ring size) uniquely influence fuel reactivity at engine relevant conditions. While the physical properties of cycloalkanes blended with Jet A have been reported in the literature, this study examines the impact of cycloalkane additives on the formation of stable intermediates during HEFA pyrolysis. Combustion studies of jet fuels have shown that larger hydrocarbon molecules undergo pyrolysis to form stable intermediates, such as methane, ethylene, and >C₂ alkenes. As these intermediates govern the oxidation of the fuel, measuring their time histories and yields provides insight into the fuel reactivity at engine relevant conditions and supports the development of combustion models. Shock tube experiments were conducted to study the pyrolysis of HEFA blends with bio-derived cycloalkanes such as 1,4 dimethylcyclooctane, p-menthane, and n-butylcyclohexane. Multiwavelength laser absorption spectroscopy (LAS) was employed to measure the time-resolved evolution of the stable pyrolysis products. All three cycloalkanes have the same carbon number, allowing for a direct comparison of how structural differences influence the formation of pyrolysis products. Blends containing 30% cycloalkanes by volume in HEFA were analyzed in experiments utilizing 1% fuel/argon test mixtures at a nominal pressure of 2 atm over the temperature range of 1150–1450 K. Additionally, ignition delay times were measured for stoichiometric mixtures for HEFA and cycloalkane blends with oxygen at a nominal pressure of 2 atm over a temperature range of 1200–1400 K. These ignition delay times were used to compare the effect of blending on global combustion behavior. These results suggest that the addition of bio-derived cycloalkanes, which improve the energy density of jet fuels, do not negatively impact the combustion performance of HEFA. Hence, the comparative performance against aromatics should ultimately guide the selection of the most suitable cycloalkane additive. Furthermore, the new measurements reported in this study can also serve as valuable targets to develop accurate chemical kinetic models.

1. Introduction

The aviation industry, which accounts for approximately 2% of global CO₂ emissions, faces mounting pressure to transition to low-carbon energy solutions while maintaining performance standards [1].

Sustainable Aviation Fuels (SAFs) have emerged as a pivotal component in achieving these environmental goals, offering the potential to significantly reduce greenhouse gas emissions and particulate matter [2].

Hydroprocessed Esters and Fatty Acids (HEFA), derived from renewable sources, are among the Sustainable Aviation Fuels (SAFs) currently

* Corresponding author.

E-mail address: alkap@stanford.edu (A. Panda).

¹ These authors contributed equally to this work.

under development. The primary feedstocks for HEFA production include vegetable oils, used cooking oils, and animal fats. Additionally, non-edible oils from crops like jatropha and camelina, as well as algae-derived oils, have been employed to diversify feedstock sources and mitigate competition with food production [3]. HEFA fuels have gained prominence as they have received ASTM certification for use in blends containing up to 50% fossil jet fuel, making them one of the most mature renewable jet fuel technology to date [3]. Despite their advantages, the limited energy density and lack of aromatic properties hinder their full deployment [4,5]. Recent combustion studies further underscore the role of composition in HEFA combustion behavior. Liang et al. showed that differences in alkane and aromatic content compared to conventional RP-3 result in stronger reactivity at low temperatures, whereas Wang et al. observed that ignition-delay times for neat HEFA and HEFA/RP-3 blends remain largely unchanged [6,7]. The results highlight the critical role of compositional effects in optimizing HEFA performance and facilitating widespread implementation.

Fortunately, bio-derived cycloalkane additives offer promising solutions to the limitations of HEFA fuels. Unlike aromatics, which contribute to increased soot formation and pose health risks, cycloalkanes provide comparable density and combustion properties while reducing environmental and health impacts. Recent advancements in biomass conversion technologies have significantly enhanced the production of high-density cycloalkanes from lignocellulosic feedstocks [8,9]. These advancements highlight the potential of bio-derived cycloalkane additives as a sustainable alternative to address the limitations of HEFA fuels.

Additionally, prior studies have demonstrated that blending cycloalkanes with conventional fuels can achieve favorable fuel physical properties, including energy density, kinematic viscosity, and freezing point [5,10]. Some cycloalkanes also exhibit o-ring swelling behavior, making them a potential aromatic substitute that complies with ASTM D7566 standards for aviation fuels [5,11,12]. However, the performance of cycloalkanes varies depending on their molecular structure. For instance, cycloalkanes with larger ring sizes have shown significant potential as aromatic substitutes for conventional Jet-A blends [5]. On the other hand, polysubstituted cycloalkanes tend to exhibit higher yield sooting indices, which may limit their applicability in certain SAF formulations [5]. C10–C12 cycloalkane blends in Jet A have been shown to enhance energy density and reduce kinematic viscosity; incorporating these cycloalkanes into HEFA could significantly improve its physical properties, thereby increasing its compatibility with existing jet fuel specifications and facilitating broader deployment as a drop-in sustainable aviation fuel [10].

Despite some promising characteristics, the chemical complexity of cycloalkane additives in conventional jet fuels pose challenges in understanding their impact on fuel reactivity. To address this gap, we examine the ignition delay times and the formation of stable intermediates during the pyrolysis of HEFA blends containing different cycloalkanes. The cycloalkanes explored in the study are *n*-butylcyclohexane (nBCH), 1,4-dimethylcyclooctane (DMCO), and 1-isopropyl-4-methylcyclohexane, also referred to as *p*-menthane (PMT). By focusing on structural variations (monosubstituted and polysubstituted ring systems, different ring sizes) while maintaining the same carbon number, this study aims to elucidate their role in fuel decomposition pathways under engine-relevant conditions. The advantage of studying cycloalkanes in HEFA blends is that it allows for the isolation of the relative impact of the cycloalkanes themselves. Conventional jet fuel consists of four molecular classes (*n*-alkanes, isoalkanes, cycloalkanes, and aromatics), while HEFA is primarily *n*- and isoalkanes. This approach enables clearer insights into how specific cycloalkane structures influence pyrolysis and combustion properties. The findings aim to contribute to the advancement of SAF technologies and inform the design of next-generation biofuels.

2. Experimental setup & methods

2.1. Shock tube

The pyrolysis experiments were conducted using the Stanford University's Kinetics Shock Tube (KST), which has an internal diameter of 14.13 cm. The shock tube consists of a 3.35 m driver section and an 8.5 m driven section, allowing precise control of test conditions. Helium was used as the driver gas along with polycarbonate diaphragms, generating test durations of approximately 2 ms. Incident shock velocities were recorded at five axial positions along the tube using piezoelectric pressure transducers, with the shock speed extrapolated to the end wall to determine the post-reflected shock conditions. The speed of the incident shock extrapolated to the end wall and the well-defined thermodynamic state of the test mixture led to $\pm 0.7\%$ uncertainty in the calculation of the post-reflected shock temperature and $\pm 1.2\%$ uncertainty in the post-reflected shock pressure. Laser diagnostics were placed at the end of the shock tube and had optical access through ZnSe substrate windows (20 mm diameter, 3 mm thick). Optical measurements were performed 2 cm away from the end wall. The pressure at the measurement plane was also recorded using a KistlerTM piezoelectric pressure transducer model 603CAA.

Ultra high purity (99.999%) helium and argon were obtained from Linde Gas. HEFA was supplied by Professor Joshua Heynes at Washington State University Tri-Cities (gas chromatograph analysis of HEFA sample listed in Table S2 of the supplementary materials), while PMT and DMCO were provided by the Naval Air Weapons Station China Lake. nBCH was purchased from Sigma-Aldrich. All fuels were used as received ($\geq 99\%$ without further purification). For each experiment, liquid fuel was injected by syringe into an evacuated stainless-steel mixing vessel that had been heated to 150 °C. The fuel was allowed to vaporize completely, after which the vessel was back-filled manometrically with the inert gas to the desired partial pressures and mechanically stirred for 45 min. Continuous stirring, low injection volume, and sufficient heating are utilized to ensure a homogeneous gas-phase mixture and minimize the effects of preferential vaporization. The resulting dilute mixture was then transferred into the shock tube, which was maintained at 100 °C to prevent fuel condensation.

2.2. Laser diagnostics

Infrared Laser absorption spectroscopy (LAS) was used to non-intrusively track the decomposition of fuel and the formation of stable intermediates. LAS allows for quantitative and time resolved speciation without the need for any calibration. The Beer Lambert equation characterizes the relationship between the transmission of a wavelength of light through the test gas $\left(\frac{I_t}{I_0}\right)_\nu$ and the number density of the test gas n . The cross-section σ describes the propensity of the absorbing species to absorb a specific frequency of light ν .

$$\alpha(\nu) = -\ln\left(\frac{I_t}{I_0}\right)_\nu = \sigma(\nu)nL \quad (1)$$

Here, $\alpha(\nu)$ is the absorbance of light with frequency ν , and L is the path length.

The methane mole fraction was monitored using two Nanoplus Interband Cascade Lasers (ICLs) operating simultaneously at wavelengths of 3.1759 μm and 3.1758 μm . The two-color online/offline diagnostic method uses a wavelength strongly absorbed by the target species (online) and a nearby wavelength with minimal absorption (offline). By comparing the absorbance at these two wavelengths, contributions from broadband interference can be accounted for. The corresponding absorption coefficients for these wavelengths, as reported by Sur et al. [13], were utilized in the analysis.

Ethylene concentrations were measured using a two-color online/offline diagnostic technique developed by Ren et al. [14]. Two CO₂ gas lasers from Access Lasers Co., operating concurrently at 10.532

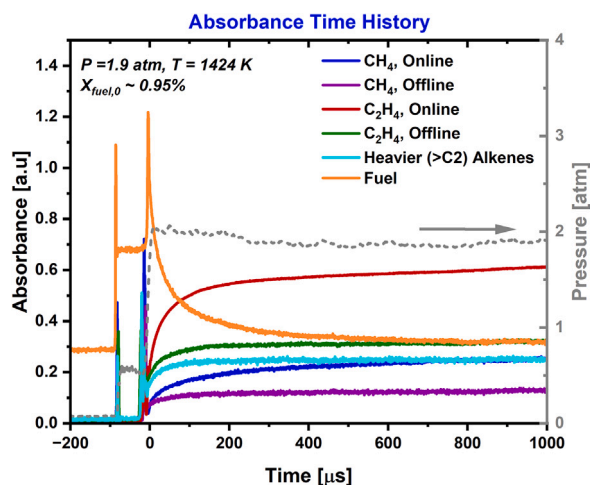


Fig. 1. Absorbance time history of six wavelengths for experiment conducted with 0.95% HEFA at 1.9 atm and 1424 K.

μm (online) and $10.675\ \mu\text{m}$ (offline), and the ethylene absorption cross-sections at these wavelengths, as reported by Ren et al. [14], were directly applied in this study.

A composite measurement of heavier alkenes ($>\text{C}_2$) was conducted using an Extended Cavity Quantum Cascade Laser (Daylight Solutions) operating at $10.962\ \mu\text{m}$. Individual heavier alkenes are challenging to distinguish spectrally due to significant overlap of absorption features. Previous studies have utilized additional spectral information in the $10\text{--}12\ \mu\text{m}$ range to estimate a composite heavier alkene mole fraction [15,16]. At this wavelength, absorption is expected to be dominated by ethylene, 1,3-butadiene, propene, and 1-butene. The absorption cross-section for these species at this wavelength is reported by Pinkowski et al. [16,17]. The ratios of individual alkenes in the product pool were determined using simulations based on JetSurF 2.0 and prior work on p-menthane GC pyrolysis [18,19]. In the absence of detailed kinetic mechanisms, a conservative assumption was made that all the heavier alkenes under consideration formed in equal proportions.

Finally, the initial fuel loading was measured using an ICL emitting at $3.41\ \mu\text{m}$ prior to every experiment. Fuel absorption cross-sections were measured using an in-house FTIR spectrometer [20]. Post-reflected shock, this will contain information about fuel decomposition and other fuel fragments but can still serve as a useful measure of overall fuel utilization.

A representative absorbance trace of 0.95% HEFA pyrolysis for all the wavelengths along with the measured pressure trace at 1424 K and 2.2 atm has been shown below in Fig. 1.

The time-resolved mole fraction measurements of stable intermediates enable a comprehensive assessment of carbon recovery during pyrolysis experiments. Fig. 2 illustrates that approximately 95% of carbon atoms were recovered from laser absorption diagnostics during HEFA pyrolysis at 1188 K. The carbon atoms were tracked through the evolution of major stable products and the parent fuel, demonstrating the robustness of the diagnostic approach.

2.3. Emission diagnostic

Ignition was identified by detecting OH^* chemiluminescence around 306 nm, corresponding to the (0,0) band of the $\text{A}^2\Sigma^+ \rightarrow \text{X}^2\Pi$ system, using a modified PDA36 A silicon detector equipped with a Schott UG11 filter. The ignition delay time was defined as the interval between the reflected shock arrival and the extrapolated zero crossing of the peak slope of the OH^* emission trace.

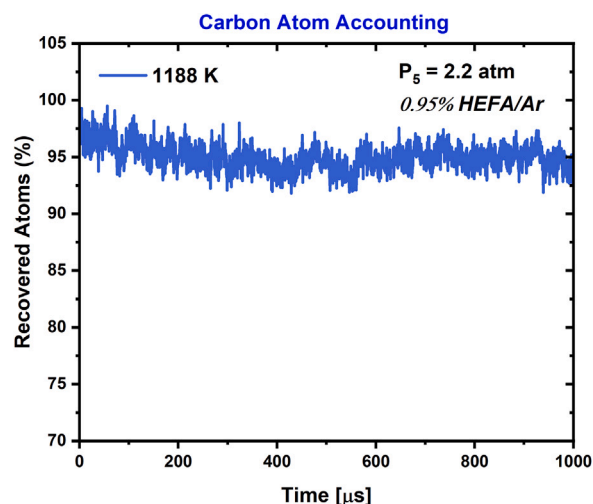


Fig. 2. Percentage of recovered carbon atoms during a representative experiment conducted in 0.95% HEFA/Ar at a nominal condition of 2.2 atm and 1188 K.

2.4. Spectral surrogate

To accurately determine the mole fractions of pyrolysis products, a temperature profile is needed in addition to the measured absorbance traces. Typically, a temperature time history is obtained through simulation of the fuel at experimental conditions. However, the absence of a detailed chemical kinetic mechanism for the HEFA fuel sample necessitated the development of a surrogate fuel. A spectral surrogate was constructed by identifying a mixture of neat hydrocarbons whose combined spectrum closely matched that of HEFA, along with matching its hydrogen-to-carbon (H/C) ratio and molecular weight (MW). The absorption spectrum for the HEFA sample was obtained using a Fourier transform infrared spectrometer (Nicolet 6700 FTIR), which provided absorbance measurements in the $2\text{--}15\ \mu\text{m}$ range. To construct the surrogate, a list of 20 previously measured relevant hydrocarbons was used as potential components for fitting. The spectrum for these hydrocarbons were obtained from measurements at Stanford as well as the PNNL spectral database [21]. These hydrocarbons were selected based on their relevance to jet fuel composition. A constrained linear least-squares optimization approach was employed to determine the mole fractions of the surrogate components. The surrogate components and the respective mole fractions have been listed in Table S3 of the supplementary materials. The resulting surrogate had a MW within 0.75% and an H/C ratio within 1.7% of the HEFA fuel sample. Fig. 3 illustrates the comparison between the measured HEFA FTIR absorption spectrum and the surrogate spectrum in the $3\ \mu\text{m}$ region.

The close agreement between the two spectra provides an optimistic validation of the constrained linear least squares optimization and affirms the surrogate's ability to replicate the spectral characteristics of the HEFA sample. As has been demonstrated, FTIR spectra serve as a reliable predictor of both physical and chemical properties of fuels. For instance, research has shown high correlations between mid-infrared FTIR absorption spectra and key properties such as density, initial boiling point, and molecular weight, among others [22,23].

Once the spectral surrogate was developed, a lumped chemical kinetic model, which contained the relevant hydrocarbons, was employed for simulations [24]. This model enabled the simulation of temperature profiles for all experimental conditions.

3. Results and discussion

3.1. Time histories of stable intermediates

The time-resolved mole fractions of key stable intermediates, including methane, ethylene, and a composite ($>\text{C}_2$) alkene group, were

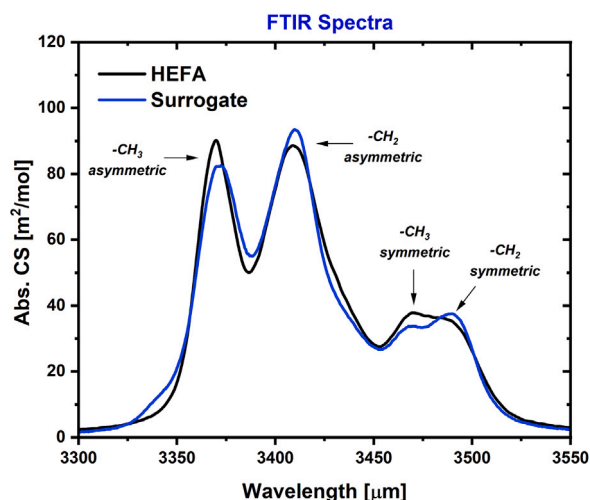


Fig. 3. Comparison between measured vapor phase absorption cross-section of HEFA and its developed surrogate.

measured during the pyrolysis of HEFA and HEFA/cycloalkane blends. These intermediates provide insight into the decomposition pathways and reactivity of the fuel under engine-relevant conditions. Experiments were conducted at nominal pressures of 2.2 atm over a temperature range of 1150–1450 K, using roughly 1% fuel/argon mixtures. The temperature-dependent time histories for HEFA and HEFA/BCH blends are presented here to illustrate how blending affects the overall profiles of the time-resolved data. Further insight into the effect of cycloalkane blending on fuel performance will be discussed in Section 3.2, where the pyrolysis yields at various temperatures are compared to highlight the structural influence of different cycloalkanes.

Fig. 4 illustrates the mole fraction time histories for the intermediate species during HEFA pyrolysis at a representative temperature of 1424 K. The shaded regions around the measured time histories represent the uncertainty in the experimental data. These uncertainties arise from multiple sources, including variations in post-shock temperature and pressure, inaccuracies in absorption cross-sections, and noise in the raw absorbance signals. The methodology for propagating these uncertainties in multi-wavelength laser absorption experiments has been rigorously analyzed in prior work, such as that by Biswas et al. [15].

The following figures, Figs. 5 and 6, display the impact of the addition of nBCH to HEFA at comparable temperature conditions. Notably, the overall shape of the time histories remains relatively similar between the figures, reflecting consistent reaction pathways. The addition of nBCH results in intermediate formation similar to those of pure HEFA, indicating that nBCH may not significantly alter the initial decomposition pathways or reduce the availability of precursors for product formation. The time-resolved speciation for all neat cycloalkane additives as well as cycloalkane-HEFA blends can be found in the supplementary materials.

3.2. Impact of molecular structure

3.2.1. Cycloalkane additives

Methane, ethylene and heavier (>C₂) alkene yields at 1 ms for the cycloalkane additives nBCH, PMT and DMCO are shown in Fig. 7. All measured yields for intermediate species are normalized with their respective initial fuel loading to remove the effect of minor differences in fuel mixtures in different experiments. Yields for all intermediates show an increase with rising temperatures as expected from their mole fractions time histories. At temperatures greater than 1400 K, ethylene yields at 1 ms begin to plateau. In the case of heavier alkenes, yields at 1 ms begin to decrease at temperatures greater than 1400 K fitting

well with the trends seen in the mole fraction time histories of heavier alkenes. This may be attributed to their decomposition into ethylene and other smaller stable intermediates.

As shown in Fig. 7(a) below, PMT and DMCO have comparable methane yields at 1 ms, while the methane yield for nBCH is approximately 50% of these values across all temperatures. Ethylene yields at 1 ms are highest for nBCH, followed by DMCO and then PMT. Additionally, the composite heavier alkene yields at 1 ms are similar for both PMT and DMCO, with nBCH yield being the lowest.

Previous shock tube studies and kinetic modeling of alkylated cycloalkanes show that pyrolysis primarily proceeds via three reaction pathways: unimolecular decomposition, ring-opening isomerization, and H-abstraction reactions. Unimolecular decomposition of fuel leads to the formation of methyl and H radicals. Methyl radicals subsequently abstract H at different sites of the fuel to produce methane. Methane is also formed from downstream reactions involving C1–C3 species. Ring-opening isomerization leads to the formation of large branched alkenyl groups that undergo successive β C–C/C–H scission to form smaller stable alkenes and dienes [25–27].

Boddapati et al.'s analysis of detailed kinetic models for alkylated cycloalkanes informs the interpretation of the measured intermediate yields in this work [25]. In the case of PMT and DMCO, unimolecular decomposition yields methyl groups that abstract H from the fuel, leading to enhanced methane yields compared to nBCH. In nBCH, the n-butyl group likely undergoes β scission and contributes to the ethylene formed during the ring-opening isomerization of the cyclohexyl group. This mechanism likely results in nBCH having the highest ethylene yield. In contrast, DMCO ring opening produces a larger alkenyl group compared to PMT, which may explain its relatively higher ethylene yield.

For PMT, unimolecular decomposition also forms isopropyl radicals. At this pressure, these isopropyl radicals are more prone to undergo β -C–H scission to produce propene and hydrogen radicals rather than isomerizing into n-propyl radicals that produce ethylene [19]. This reaction pathway boosts the heavier alkene yields for PMT. The ring-opening isomerization of cyclohexyl groups in both PMT and nBCH likely leads to similar proportions of ethylene and larger alkenes such as propene and 1,3-butadiene.

In the case of DMCO, the absence of detailed kinetic mechanisms necessitated the assumption of an equal proportion of propene, 1,3-butadiene, and 1-butene in the product distribution. This assumption aligns the alkene yields at 1 ms with those of PMT. However, it is plausible that the ring-opening isomerization step in DMCO produces a much larger alkenyl group compared to PMT and nBCH. These larger alkenyl groups can further break down into larger C5–C6 alkenes and dienes. While these larger alkenes eventually decompose into smaller species like ethylene and propene, this might not occur within the experimental time scale. Additionally, larger alkenes have greater cross-sections, and when smaller cross-sections of C3–C4 alkenes are attributed to the signal, the reported alkene yields may be inflated.

This detailed understanding of the pyrolysis pathways and their intermediates provides insights into the differences in product distributions among PMT, DMCO, and nBCH.

3.2.2. Impact of blending on HEFA combustion

Methane, ethylene, and heavier alkene yields at 1 ms for pure HEFA and volumetric blends of various cycloalkane additives with HEFA are presented in Fig. 8. Similar to the trends observed with pure cycloalkanes, increasing temperatures lead to enhanced fuel reactivity and higher yields for all intermediates.

Although the scatter-free trends in the 1 ms yields reveal minor differences attributable to the distinct molecular structures of the additives, they largely mirror the trends observed in the pyrolysis products of pure cycloalkanes. The most notable feature in these results is the pronounced influence of the multi-component nature of HEFA in these blends. Despite the significant proportions of cycloalkanes added to the

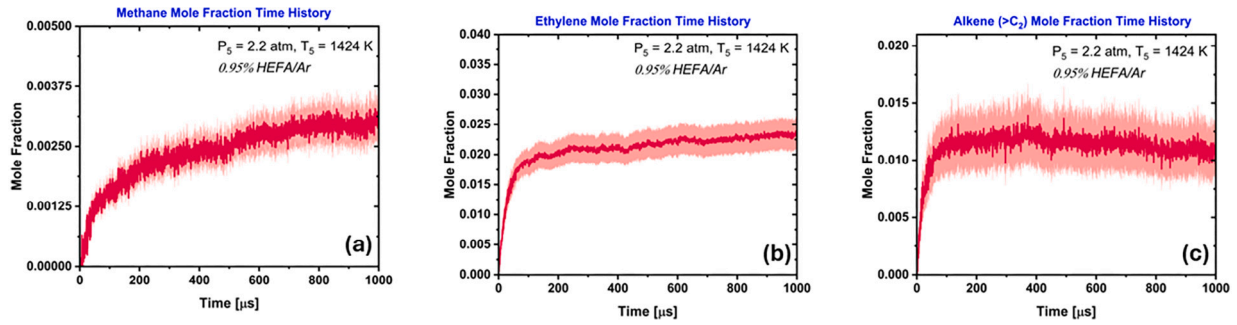


Fig. 4. Mole fraction time histories and uncertainties for (a) methane, (b) ethylene, and (c) alkenes (>C₂), shown for an experiment conducted in 0.95% HEFA/Ar at 1424 K.

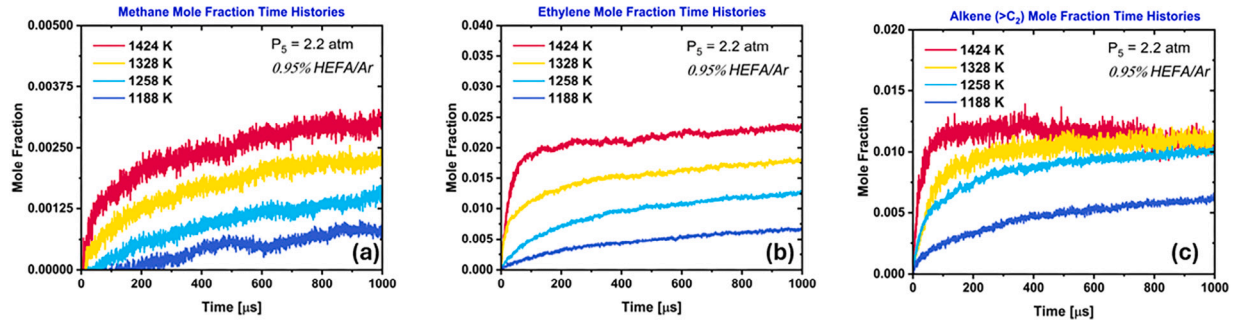


Fig. 5. Comparison of mole fraction time histories of (a) methane, (b) ethylene, and (c) alkenes (>C₂) at different temperatures, measured in 0.95% HEFA/Ar mixtures.

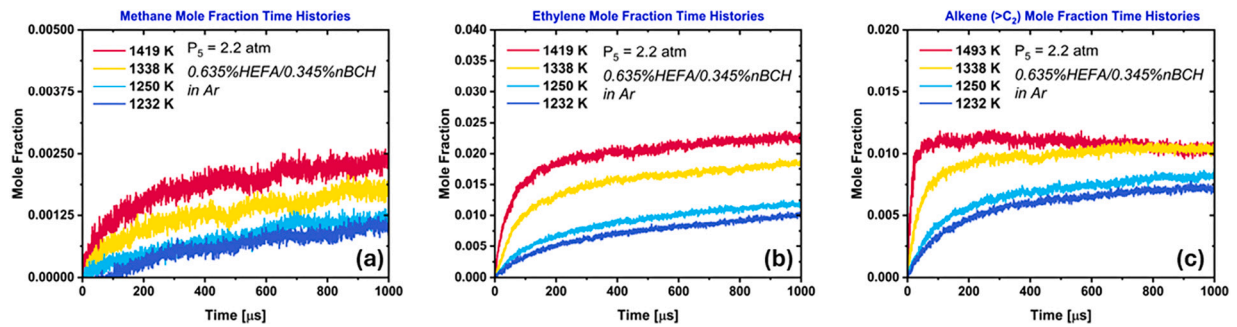


Fig. 6. Comparison of mole fraction time histories of (a) methane, (b) ethylene, and (c) alkenes (>C₂) at different temperatures, measured in 0.635% HEFA, 0.345% nBCH in Ar mixtures.

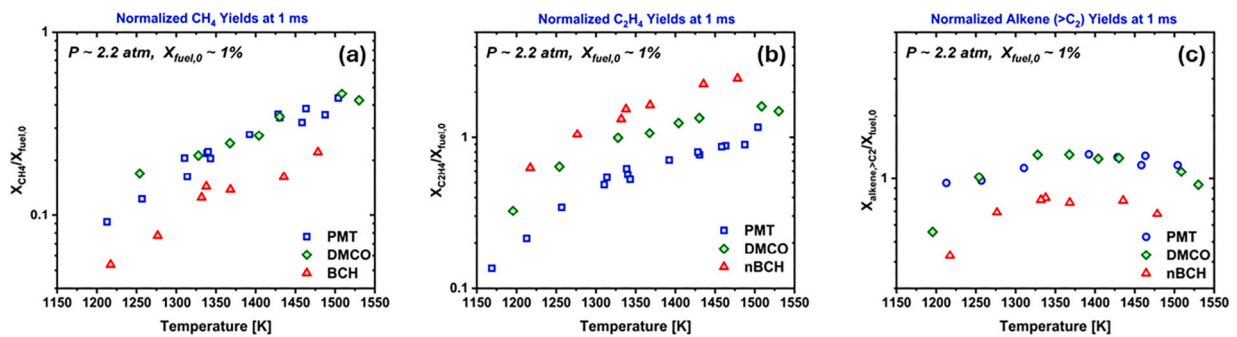


Fig. 7. Comparison of normalized yields at 1 ms of (a) methane, (b) ethylene, and (c) alkenes (>C₂) at different temperatures, measured in 1% cycloalkane/Ar mixtures.

blends, the differences in intermediate yields at 1 ms are effectively minimized, with all yields converging toward those of pure HEFA. This highlights the dominant role of multi-component fuels in governing the overall reaction pathways and product distribution in these blends.

To assess the impact of cycloalkane blending on global combustion behavior, ignition delay times (IDTs) were measured for HEFA and its 30% cycloalkane volumetric blends. Shock tube experiments were

conducted using stoichiometric fuel/oxygen mixtures in argon at a nominal pressure of 2 atm over the temperature range of 1200–1400 K. As shown in Fig. 9, the measured ignition delay times for all HEFA-cycloalkane blends closely follow those of pure HEFA across the entire temperature range. This indicates that the addition of cycloalkanes does not significantly alter the global reactivity of the fuel mixtures under high-temperature conditions. The similarity in IDTs suggests that

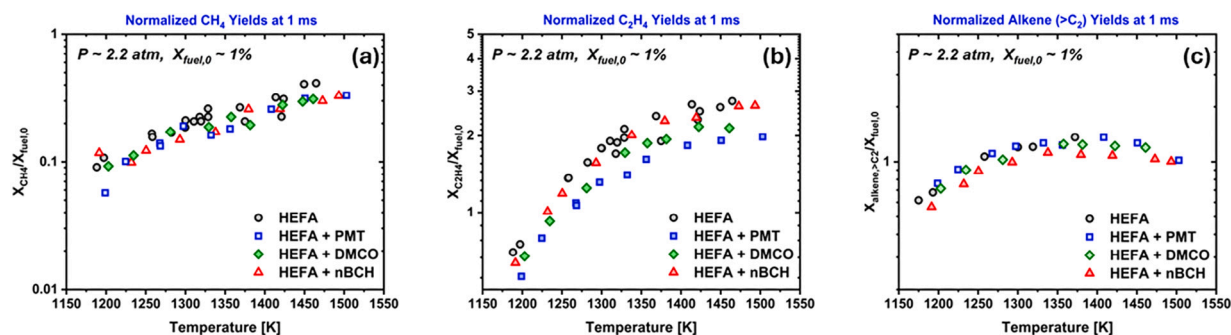


Fig. 8. Comparison of normalized yields at 1 ms of (a) methane, (b) ethylene, and (c) alkenes (>C₂) at different temperatures, measured in 1% HEFA-cycloalkane/Ar blend mixtures.

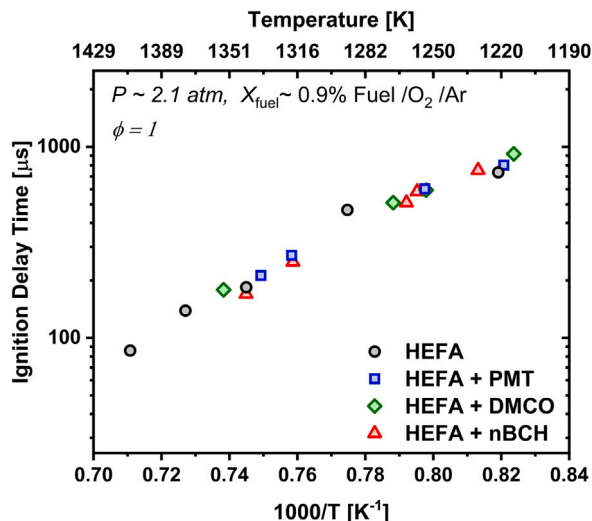


Fig. 9. Comparison of ignition delay times of 0.9% HEFA and HEFA-cycloalkane blends (30% by vol.) in stoichiometric mixtures with oxygen and argon at a nominal pressure of 2.1 atm.

the dominant ignition chemistry is governed by the HEFA base fuel, and the structural differences among the cycloalkane additives do not appreciably affect the overall ignition behavior.

4. Conclusions

This study investigated the impact of three C₁₀ cycloalkane additives (nBCH, PMT, DMCO) on the formation of stable intermediates during the pyrolysis of HEFA fuel as well as their global combustion behavior under engine-relevant conditions. Time-resolved mole fraction measurements of key intermediates, including methane, ethylene, and heavier alkenes, were conducted using multiwavelength laser absorption spectroscopy in a shock tube. IDTs were measured using OH* emission at 306 nm.

A comparative analysis of the pyrolysis yields and IDTs underscored the influence of structural variations, such as ring size and degree of substitution, on decomposition kinetics. The results demonstrate that the intermediate yields and IDTs of the HEFA/cycloalkane blends closely resemble those of pure HEFA, highlighting the dominant role of HEFA in determining reaction pathways. This convergence emphasizes the multi-component nature of HEFA, which effectively minimizes the impact of added cycloalkanes on product distribution. This outcome is particularly promising, as it suggests that HEFA's favorable combustion characteristics are preserved even with the inclusion of high energy density cycloalkane additives. The ability to improve fuel physical properties without compromising combustion performance enhances the practicality of HEFA as a flexible base fuel for SAF development.

Assessing the impact of cycloalkane structure on product formation and ignition behavior represents a pivotal step toward understanding the high-temperature reactivity of novel SAF candidates. Future studies should include comparisons with other molecular classes, such as aromatic substitutes, to further elucidate structure-reactivity relationships.

Novelty and significance statement

This study presents a systematic investigation into how structural variations among bio-derived cycloalkanes affect the pyrolysis behavior and combustion characteristics of HEFA jet fuel under engine-relevant conditions. Using multiwavelength laser absorption diagnostics in a shock tube, we provide new time-resolved measurements of stable intermediates for HEFA blends containing three structurally distinct cycloalkanes with identical carbon numbers. The novelty lies in isolating the influence of molecular structure on intermediate formation by studying blends with a consistent base fuel. The results show that despite structural differences among additives, the pyrolysis product distributions and ignition delay times of the blends closely resemble those of HEFA. This finding is significant because it demonstrates that physical property enhancements from cycloalkane addition can be achieved without compromising combustion performance. These insights support the development of cleaner, high-performance SAFs.

CRediT authorship contribution statement

Alka Panda: Writing – original draft, Writing – review & editing, Designed research, Performed research, Formal analysis. **Andrew Klingberg:** Writing – original draft, Writing – review & editing, Designed research, Performed research, Formal analysis. **Ronald K. Hanson:** Writing – review & editing, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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This research was partially supported by the Naval Air Weapons Station China Lake. Any opinions, findings, conclusions, or recommendations expressed in this material are those of the authors and do not reflect the views of the Naval Air Weapons Station China Lake.

Appendix A. Supplementary data

GC analysis of the HEFA sample, details of the surrogate fuel components, and time-resolved speciation data from all experiments are included in the Supplementary Materials in the online version of the publication.

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.proci.2025.105803>.

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